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Optimising ESP Runlife – A Practical Checklist

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Abstract

Optimising ESP runlife is essential to avoid additional workover costs and maintain production levels. Although some failures may require specialist equipment or solutions, the vast majority of short runlives are preventable. These problems usually result from a lack of consideration of the entire ESP system during the design phase or shortcomings during the installation and operations phases.

This paper will examine the most common causes of premature ESP system failure and identify the reasons and possible solutions for the problem. A checklist has been devised to assist new or existing users of ESPs to design, install and operate an ESP system to achieve an optimum runlife for particular well conditions.

Introduction

In many areas of the world where ESPs are being newly introduced a common concern among users is the potential for premature failures, thus rendering the economic benefit of ESPs vs. other lift methods (higher production rates) null and void. Many companies will avoid the use of ESPs until all other possibilities are exhausted, primarily due to this fear of the unknown and perceived unreliability of ESP systems.

In fact, the volume of experience with ESPs in a huge variety of applications and conditions is such that most problems have been identified and are readily fixed. In other words, the “learning curve” has already been learnt and there is no reason that this expertise should not widely applied - especially for the benefit of new users.

Most failures of ESPs are electrical in nature, since this is the lifeline that provides energy to the motor and pump and is often the weakest link of the system. However, the real cause of failure could be elsewhere (such as reservoir inflow plugging or a valve shut at surface causing motor and pump heating) and a proper cause and effect analysis should always be conducted.

Common failure modes and causes are noted below, together with relevant solutions and a discussion of the merits of taking an integrated approach to optimising ESP system runlife. The use of a practical checklist will be reviewed to show how virtually all issues and problems that effect runlife have easily implemented solutions.

Background

An ESP system is made up of five principal components:

- 1) a multistage centrifugal pump consisting of rotating impellers and fixed diffusers
- 2) a high speed (≈ 3500 rpm) three-phase electric motor filled with dielectric oil
- 3) a seal section which isolates well fluids from the motor and accommodates shaft movement
- 4) a three conductor electric cable which runs through the wellhead down to the motor
- 5) a power supply system at surface (generator, transformer, variable frequency drive)

Problems can therefore occur in either the pump, seal/protector section or electrical (cable/motor) system. A schematic illustrating the whole system is given in Attachment 1.

The pump

The pump generates head by the conversion of energy in two steps:

- the impeller imparts shaft energy (mechanical kinetic) to the fluid (hydraulic kinetic)
- the diffuser converts fluid energy (hydraulic kinetic) to potential energy

The potential energy (or head) generated depends on the following:

- speed of rotation of the impellers (**rpm**)²
- **flowrate** of fluids through the impeller
- **size** (diameter) and **shape** (radial, mixed, axial) of stage

The efficiency of energy conversion is also effected by:

- the amount of **gas** flowing through the impeller
- the **viscosity** of fluids through the impeller
- other mechanical factors such as **wear**

The seal/protector section

The seal or protector section has two main functions:

- to balance pressures and isolate fluid movement between the wellbore and motor
- to isolate shaft movement between the pump and the motor

This is accomplished by:

- bag or labyrinth seals (the latter relies on a manometer principle, ineffective at high angles)
- thrust bearing pads to accommodate downthrust or upthrust forces from the pump

The electrical system

The electrical system must provide high power (up to 800 hP) to the motor as follows:

- high volts (up to 5000 v) and current (up to 180 A) at a variable frequency (30 – 70 Hz)
- this power must be isolated from ground in all components (cable, connectors, splices etc.)
- the power supply must be free of harmonics or spikes (noise)

This is accomplished by:

- generators and transformers to provide the required output voltage
- variable frequency drives which convert a base input frequency to the required output freq.

The Problems

Failure modes in ESP systems result from one or a combination of factors that compromise the operation of each component. Since the principles of operation of each component is known, then the causes of failure in each can easily be identified:

The pump

The high fluid velocities through the impeller vanes in a pump stage give rise to several problems related to frictional pressure losses:

- **gas** bubbles greater than the impeller vane size results in cavitation and gas-locking
- high **viscosity** fluids lower efficiency of energy conversion due to higher friction
- **sand** or **scale** causes erosional wear and/or plugging of the impeller vanes and pump intake
- any wellbore **debris** sucked into the pump will result in blocking or damage to the stages

Operation out of the range of the pump (too little or too much **flow**) also causes problems due to low efficiency (giving rise to **heating**) and mechanical wear between the impellers and diffusers.

Initial sizing of the pump is done by determination of average flowrate in the pump (rb/d) and total dynamic head (TDH) required. These depend primarily on the well inflow performance (productivity index and reservoir pressure) and fluid properties (mainly density and viscosity). If either of these **design input assumptions** is incorrect, then the pump design is wrongly specified and early failure could result.

The seal/protector section

The seal section is generally quite robust, but failure could occur due to the following:

- very high shaft **load** caused by operation out of range of pump (up/downthrust)
- high shaft **vibration** due to debris or wear in pump
- operation at extreme **temperature** or fast **pressure** changes resulting in seal failure
- operation at high angles with labyrinth type seals

The electrical system

Since the electrical system is simply a series circuit, any component that is the weakest link will result in system failure. Common problems include:

- **damage** to any component due transportation, handling or installation
- extreme **heating** of the cable or motor (due to low flow, operation out of pump range etc.)
- lack of heat transfer medium around the motor (i.e. **gas** build-up)
- **high shaft thrust** or loading due to a pump problem (out of range, gas cavitation, wear etc.)
- operation with **unclean power supply** (i.e. voltage spiking or fluctuations, harmonics etc.)
- use of a single **inferior component** in any part of the electrical system

Operational problems can also give rise to premature electrical failures, e.g. shut-in of surface or sub-surface safety valve with the ESP running, plugging of perforations or pump intake, downhole recirculation etc.

The Solutions

A systematic examination of each of the failures modes above leads to the following solutions that either eliminate or accommodate the problem through proper equipment selection or improved installation procedures and operations practices. Some examples for the most common issues are given below – a full set of solutions is given in the attached **checklist**.

Gas handling

Gas is a very common issue with ESP systems, but problems can be avoided by consideration of the following during the **design** phase:

- set the pump as **deep** as possible (to raise pump intake pressures)
- use large, **mixed flow** impellers (bigger vane size than radial flow) – up to 30% free gas
- install a **gas handler** (no venting required) – up to 60% free gas at intake
- install a **gas separator** (requiring venting up the annulus) – up to 80% free gas at intake
- use **axial flow** impellers (requires use of hydraulic submersible pump) – up to 80% free gas
- avoid **tight clearances** around motor (to avoid annular gas build-up)

During the **operational** phase, i.e. once the ESP has been run, consider the following:

- increase wellhead pressure by choking the well (raises intake pressure)
- lower ESP supply frequency (raises intake pressure)

High viscosity fluids

High viscosity oils or emulsions can be handled by the following **design** phase considerations:

- use large, **mixed flow** impellers (bigger vane size than radial flow) – less friction
- design for **lower rpm** with more stages – lowers friction
- add means to **inject demulsifier** at pump intake (e.g. capillary line)
- for extremely high viscosities, use a **progressive cavity pump** (PCP)

During the **operational** phase, consider the following:

- lower ESP supply frequency (to lower rpm and hence friction)
- inject demulsifier at pump intake

Sand, scale or debris

These are common contaminants of ESP systems and can be handled during the **design, installation** and **operational** phases as follows:

- prevent sand influx in the first place, by use of a **gravel pack** or **screens**
- **inject scale inhibitor** at pump intake or periodically scale squeeze
- **clean out wellbore** prior to ESP installation
- use large, **mixed flow** impellers (bigger vane size than radial flow)
- start up pump **slowly** to prevent ingestion of sand/debris
- use a special **pump design** for sand (upgraded metallurgy, radial wear protection etc.)

The electrical system

As the electrical system is the most common immediate cause of failure, particular attention should be paid to the following:

- ensure a **clean power supply** system (especially if using variable frequency drives)
- use **top quality components** throughout the electrical system
- upgrade **installation procedures** (time for splice, slow run in hole, use cable clamps etc)

An Integrated Approach

When optimising the reliability and hence runlife of ESP systems, it is essential to take a whole system or integrated approach. This will avoid overlooking any element that could be the weak link of the system that causes premature failure. Some guidelines to this approach are given below.

1) An ESP should always be considered as part of the whole well system:

- includes reservoir inflow, fluid properties, ESP, completion and electrical components
- each should be considered both separately AND integrated with the other system elements
- failure of a single component usually comprises whole system integrity
- three major system components; mechanical, hydraulic and electrical:

<i>Mechanical</i>	Packer, tubing size and material, bypass, corrosion, scale, dual ESPs, casing ID, dogleg severity, deviation, seal
<i>Hydraulic</i>	Reservoir performance (P_R and PI), sand, scale, fluid properties, gas, solids deposition (wax, asphaltenes), watercut development, flow correlations, pump, monitoring
<i>Electrical</i>	Power supply quality, transformer tapplings, variable speed drives, harmonics, filters, cable, wellhead and packer penetrators, splices, pothead, motor, monitoring

2) Always consider the ESP life cycle (rather than just one phase):

- three phases during ESP life; design, installation, operations
- a flaw in any one of these phases can comprise ESP runlife
- consider the following phases:

<i>Design</i>	Reservoir, drilling, completion, electrical, surface facilities, fluid properties, previous experience (own company, other operators, pump manufacturers), innovation, economics
<i>Installation</i>	Rig crew training, working environment, procedures and practices, patience, start-up, training, training, training!
<i>Operations</i>	Field personnel training, monitoring, intelligent control systems, well/ESP performance analysis and interpretation

A **practical checklist** that identifies problems in each of the three system components (hydraulic, mechanical and electrical) and how solutions can be implemented in each of the life cycle phases (design, installation and operation) is attached.

3) During the operations phase, continuous downhole monitoring is essential:

- set alarms and trips on critical variables
- analyse and interpret data to understand well, reservoir and ESP performance

<i>Parameters</i>	<i>Control/Optimisation Function</i>
Pump discharge pressure	Well shut-in protection
Wellhead & pump discharge pressure	Watercut or tubing GOR estimation
Pump intake pressure	Gas locking (bubble point, free gas volumes) Reservoir drawdown (sand control)
Pump dP	Upthrust and downthrust protection; flow estimation Pump performance (wear, viscosity, manufacture)
Motor temperature	Overheating (overload, lack of cooling)
Motor amps	Overload/underload protection
Current leakage	Overheating (insulation breakdown)

A fully instrumented ESP system with appropriate alarms and trips set can prevent many of the most common failure modes, without operator intervention. More subtle problems require careful analysis and interpretation of all parameters (e.g. determination of viscosity effect on pump performance).

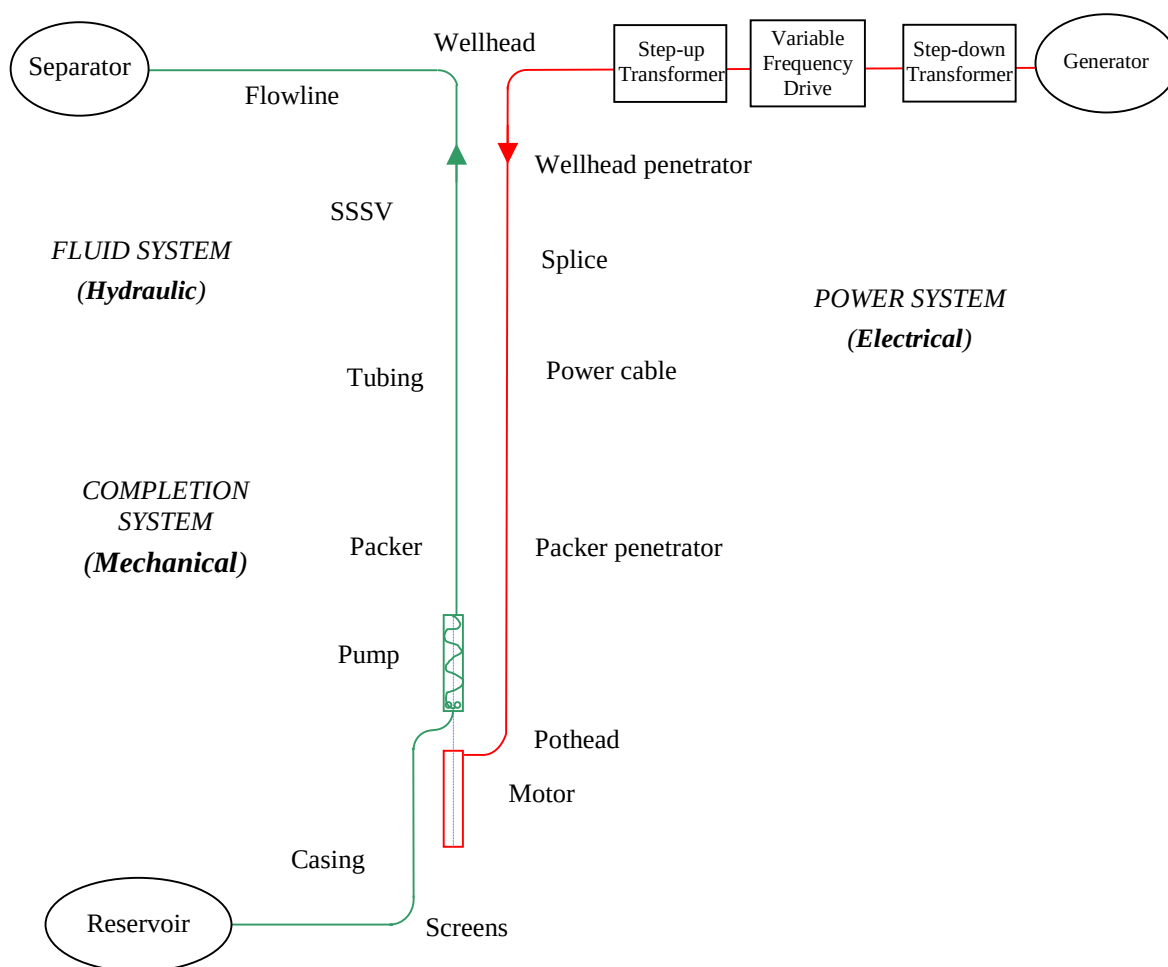
Conclusions

ESPs are a reliable, high-rate artificial lift system. Most failures are preventable with readily identifiable causes. These failures can be avoided by following a systematic, integrated approach to the entire ESP system in the design, installation and operational phases of a field development.

Monitoring of an ESP system is essential to protect against obvious failure modes and to provide critical data to allow more complex problems to be solved. A problem cannot be fixed unless it is first identified and the cause determined.

Use of the practical checklist will ensure that most of the key points effecting ESP runlife are considered, thus eliminating premature failures and optimising ESP system runlife.

The Integrated ESP System



System Components

<i>Mechanical</i>	Packer, tubing size and material, bypass, corrosion, scale, dual ESPs, casing ID, dogleg severity, deviation, seal
<i>Hydraulic</i>	Reservoir performance (P_R and PI), sand, scale, fluid properties, gas, solids deposition (wax, asphaltenes), watercut development, flow correlations, pump, monitoring
<i>Electrical</i>	Power supply quality, transformer tapplings, variable speed drives, harmonics, filters, cable, wellhead and packer penetrators, splices, pothead, motor, monitoring

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	PROBLEM	SOLUTION		
		DESIGN	INSTALLATION	OPERATION
H Y D R A U L I C	☐ Gassy well (high free gas at intake)	- Gas separator or handler - Use large mixed flow impellers - Lower drawdown (higher PIP) - Deeper pump setting depth (higher PIP) - Avoid tight clearances around motor - Use axial flow impellers (HSP)	-	- Increase wellhead pressure (choke) - Decrease frequency - Monitor to determine effects
	☐ Scale/Corrosion (H ₂ S, CO ₂)	- Inject inhibitors - Formation scale squeeze - Upgrade metallurgy, monel coating	-	- Backflush through pump - Acid soak pump (for CaCO₃) - Monitor to determine effects
	☐ High viscosity, emulsions	- Use large mixed flow impellers - Design for lower rpm, more stages - Inject demulsifier at pump intake - Use displacement pump (PCP)	-	- Increase wellhead pressure (choke) - Decrease frequency - Monitor to determine effects
	☐ Wax/Asphaltene	- Inject inhibitors - Use large mixed flow impellers	-	- Xylene backflush or soak pump - Monitor to determine effects
	☐ Pumping off (PIP to zero)	- Reduce number of stages - Stimulate reservoir	-	- Increase wellhead pressure (choke) - Decrease frequency
	☐ High temperature (>250 deg F)	- Upgrade equipment (high temp. ratings) - De-rate motor design	-	- Vigilant monitoring & control - Avoid up/downthrust operation
	☐ Unknown reservoir performance (PI)	- Estimate PI from DSTs, Darcy inflow - Use compression pump or inserts - Use variable frequency drive	-	- Increase or decrease choke - Increase or decrease frequency - Monitor to determine PI for next ESP
M E C H A N I C A L	☐ Sand production/erosion/wear	- Install gravel pack or screens - Use large mixed flow impellers - Upgrade pump metallurgy & design - Design for lower rpm, more stages - Lower drawdown (higher PIP) - Use compression pump or inserts	-	- Increase wellhead pressure (choke) - Decrease frequency - Monitor to determine effects
	☐ Subsea or remote location	Install dual ESPs for back-up	-	Vigilant monitoring & control
	☐ Damaged/weak casing	Run liner or patch	-	Limit drawdown
	☐ Debris/junk in well	Improve workover procedures	Clean up well bore	Minimise drawdown on start-up
	☐ High deviation	- Use bag type seal/protector section - Use centralisers	-	-
	☐ Doglegs	- Minimise doglegs during drilling - Design for higher rpm, less stages - Use large OD (shorter) pump	- Drift wellbore - Slow RIH	-
E L E C T R I C A L	☐ Poor power supply quality	- Install harmonic filters - Upgrade generators - Upgrade variable frequency drive - Use surge protectors - Install larger cable - Power system audit	-	- Monitor amps & voltage - Perform waveform analysis
	☐ Cable/splice/connector problems	- Address power supply quality issues - Improve workover procedures - Upgrade components	- Handling procedures - Use clamps	-
	☐ Very high power (>1000 hP)	- Install dual (boost) ESPs - Clean power supply essential	-	Vigilant monitoring & control
PEOPLE	☐ Operator error	Training (reservoir/drilling engs, PTs)	Training (rig crew)	Training (field operations staff)